

A

$$1. (a) \quad \frac{9}{18} = \frac{1}{2}$$

$$(b) \quad \frac{8}{18} = \frac{4}{9}$$

$$(c) \quad \frac{9}{18} + \frac{8}{18} - \frac{4}{18} = \frac{13}{18}$$

$$(d) \quad \frac{4}{8} = \frac{1}{2}$$

$$(e) \quad P(\text{red} | \text{even}) = \frac{4}{8} = \frac{9}{18} = P(\text{red}), \quad \text{so they are independent}$$

$$\begin{aligned} 2. (a) \quad P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= 0.6 + 0.79 - 0.49 \\ &= \underline{0.9} \end{aligned}$$

$$\begin{aligned} (b) \quad P(\text{neither } A \text{ nor } B) &= P((A \text{ or } B)^c) = 1 - P(A \text{ or } B) \\ &= 1 - 0.9 = \underline{0.1} \end{aligned}$$